

Quiz 1

● Graded

Student

Myles Sartor

Total Points

7 / 10 pts

Question 1

Problem 1

3 / 3 pts

✓ - 0 pts Correct

- 1 pt $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.6$

- 1 pt $P(A \cap B) = P(A) - P(A \cap B') = 0.3$

- 1 pt $P(A' \cap B') = P(S - (A \cup B)) = 1 - 0.6 = 0.4$

- 0.5 pts calculation error

- 0.5 pts Correct formulation on a and c

- 0.5 pts calculation in a but correct formulation and calculation for b and c

Question 2

Problem

4 / 4 pts

✓ - 0 pts Correct

- 2 pts incorrect maximum

- 1 pt missing work/justification for minimum

- 4 pts incorrect min and max

- 2 pts switched answers

- 2 pts incorrect min

Question 3

Problem 3

0 / 3 pts

- 0 pts Correct

- 1 pt (a) Identifying correct region

- 1 pt (a) Derivation of answer

- 1 pt (b)

✓ - 3 pts Incorrect

- 0.5 pts (b) Answer should be '0'

- 0.5 pts (b) show work / Incorrect reasoning

Quiz 1

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Problem 1. Suppose that $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B') = 0.1$. Find

(a) $P(A \cup B)$ (b) $P(A \cap B)$ (c) $P(A' \cap B')$.

a) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$0.4 + 0.5 - 0.1 = 0.8 = P(A \cup B)$$

b) $P(A \cap B) =$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.8 = 0.4 + 0.5 - P(A \cap B)$$

$$P(A \cap B) = 0.1$$

c) $P(A' \cap B') = P(A \cup B)' = 1 - P(A \cup B)$

$$1 - 0.8 = 0.2$$

$$\hookrightarrow P(A - B)$$

$$1 - P(B) = P(B')$$

$$P(A \cap B') = P(A) - P(A \cap B)$$

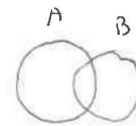
$$\hookrightarrow 0.4 - 0.1 = 0.3$$

$$1 - 0.1 = 0.9$$

$$P(B - A) \cap (A - B)$$

$$P(B \cap A')$$

$$P(A) + P(B) + P(A \cap B) - P(A \cup B)$$



Problem 2. Let $P(E)=0.9$ and $P(F)=0.8$. Find the minimum and maximum value for $P(E \cap F)$.

$$P(E \cup F) = P(E) + P(F) - P(E \cap F) \leq 1$$

$$0.9 + 0.8 - P(E \cap F) \leq 1$$

$$1.7 - P(E \cap F) \leq 1$$

$$0.7 \leq P(E \cap F)$$

Minimum value: 0.7

Maximum value: 0.8, since $F \subseteq E$, and physically

can't be greater than E
as a result of being a
subset.

Problem 3. $S = \{(x, y) : 0 \leq x, y \leq 3\}$. $y = 2 + x$
 $y = x + 2$

(a) Let $E = \{(x, y) \in S : y - x \leq 2\}$. Find $P(E)$.

(d) Let $F = \{(x, y) \in S : y = x - 1\}$. Find $P(F)$.

$$\text{Area}(S) = 3$$

$$\begin{aligned} \text{a) } P(E) &= \frac{\text{Area}(E)}{\text{Area}(S)} \rightarrow \frac{-\frac{1}{2} \cdot \frac{2}{3}}{\frac{1}{1}} = -\frac{1}{2} \cdot \frac{1}{3} = -\frac{1}{6} \\ |-\frac{1}{6}| &= \frac{1}{6} \checkmark \end{aligned}$$

$$\text{b) } P(F) = \frac{\text{Area}(F)}{\text{Area}(S)}$$

$$|4 \cdot -1| = 4$$

$$\frac{4}{3} = P(F)$$

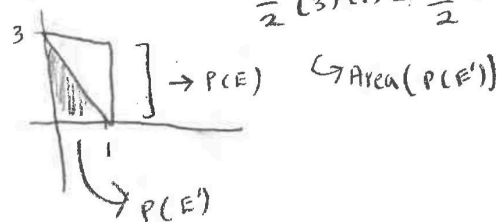
$$3 - x \leq 2$$

$$-x \leq -1$$

$$x \geq 1$$

$$\frac{1}{2}bh$$

$$\frac{1}{2}(3)(1) = \frac{3}{2}$$



$$1 - \frac{3}{2}$$

$$\hookrightarrow \frac{2}{2} - \frac{3}{2} = -\frac{1}{2} = \text{Area}(P(E))$$

$$\begin{aligned} y &= x - 1 \rightarrow y + 1 = x \\ 3 &= x - 1 \\ 4 &= x \end{aligned}$$

